Specialization, risk, and causal reasoning

A short toolkit for the ECON 342 final project

Part 1: Ma and Ramondo's paper and a Canadian decision.

Part 2: counterfactuals and differences-in-differences.

This new material will not be tested on Midterm 2.
It supports your AI-enhanced analysis and final project.

A productive cluster can also expose a household to risk

People choose a skill, industry, and place **before** learning which activities will have a good year.

1. A productive local industry attracts workers.
2. A larger cluster can become more productive.
3. A household values the average reward and the risk of a bad outcome.

In the paper's baseline, workers stay in their chosen industry and place after the shock. Wages adjust; households have no saving or financial insurance. Trade opens markets and alternative sources of supply. Its effect on **real income risk** depends on wages and consumer prices together.

Source: Ma and Ramondo, NBER Working Paper 35942, October 2026, pp. 6–15.

A higher average can come with a less attractive gamble

Invented real-consumption units. Each state has probability one-half.

Plan	Good	Bad	Average	Certainty equivalent
Specialized	120	60	90	80
Stable	84	84	84	84
Specialized, assumed 10% gain	132	66	99	88

The **certainty equivalent** is the sure amount this person values as highly as the gamble. It depends on preferences.

$$\underbrace{CE = \frac{1}{0.5/c_{\text{good}} + 0.5/c_{\text{bad}}}}_{\text{one illustrative risk preference}}$$

The stable plan beats the first gamble. An *assumed* gain in both states reverses that choice.

Source: Classroom illustration, not Canadian data or a calibrated result. The 10% gain is imposed on real consumption.

Local external scale and a firm's fixed cost

Our earlier firm example

One firm spreads a design or setup bill over more output.

$$AC = c + F/q$$

A larger run lowers fixed cost per unit.

This paper's mechanism

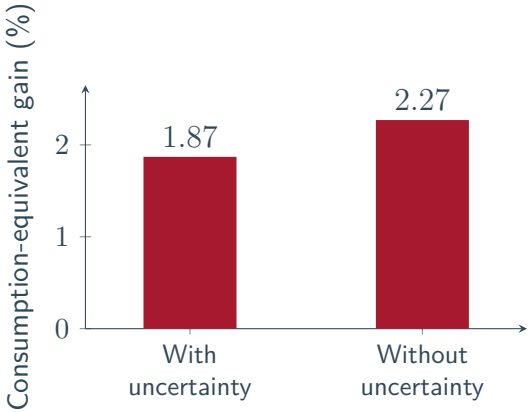
A larger industry *in one place* raises local industry productivity. Shared skills,

suppliers, and knowledge help us picture this external scale effect.

Risk can discourage workers from choosing a cluster in advance. A smaller cluster can then lose some productivity benefits of scale.

Source: Ma and Ramondo, pp. 8–13. The strength of this feedback depends on the model's assumptions and parameters.

Uncertainty reduces the modeled gain from trade

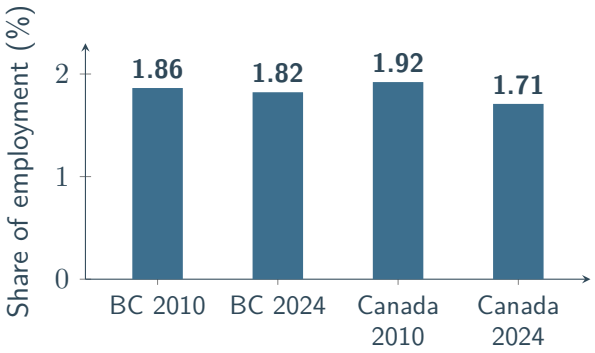


0.40 percentage points

About 18% less than the gain without uncertainty. These are **U.S. model comparisons with autarky**, expressed in welfare units. Every U.S. commuting zone gains in the latest baseline. Ideal national risk sharing largely removes uncertainty's aggregate effect.

Source: Ma and Ramondo, October 2026, Tables 6, 7, and 10. These numbers are neither Canadian estimates nor measured GDP growth.

BC resource employment grew while its share stayed similar



BC's published resource-group employment rose from **41,400 to 53,100 people**. Total employment also grew. The group's share was roughly 1.8–1.9% in both years. This describes the industrial mix. It does not establish a productivity benefit, export advantage, or causal effect.

Source: Statistics Canada 14-10-0023-01, annual employment, ages 15+, total gender. Resource group combines forestry, fishing, mining, quarrying, oil and gas. Denominator: sum of the 16 non-overlapping course groups. Rounded estimates.

The Canadian decision for your project

Choose one province and a clearly defined trade-exposed cluster. Compare it with two provinces and Canada.

- Should the province deepen a specialization, develop a feasible adjacent activity, or preserve the cluster while protecting workers from shocks?
- Exhibit 1: the industrial mix and actual trade exposure.
- Exhibit 2: historical variability, baseline and reweighted mixes, and sensitivity checks.

Employment shares describe concentration. Historical variation describes past outcomes. Add evidence on skills, suppliers, wages, prices, or adjustment to explain the welfare mechanism. **Your conclusion should be conditional:** name the evidence that would change your recommendation.

Source: Project handbook and starter data are linked on the final slide. This toolkit adds no deadline or assessment requirement.

Part 2
Causal reasoning

What would have happened without the policy or shock?

Differences-in-differences supplies one possible comparison. Its credibility comes from the design and assumptions.

For the final project. This new material will not be tested on Midterm 2.
This workshop is separate from the podcast and from the paper's model.

The missing counterfactual

A Canadian industry's employment falls after a trade restriction.

Observed	**Missing**
The industry's outcome after the restriction.	The same industry's outcome at the same time if the restriction had not occurred.

A before-and-after change also includes other forces: demand, commodity prices, technology, and the business cycle. A comparison group can help approximate the missing path. We must explain why its change is informative.

Differences-in-differences with four numbers

Hypothetical employment rates. These are invented groups and outcomes.

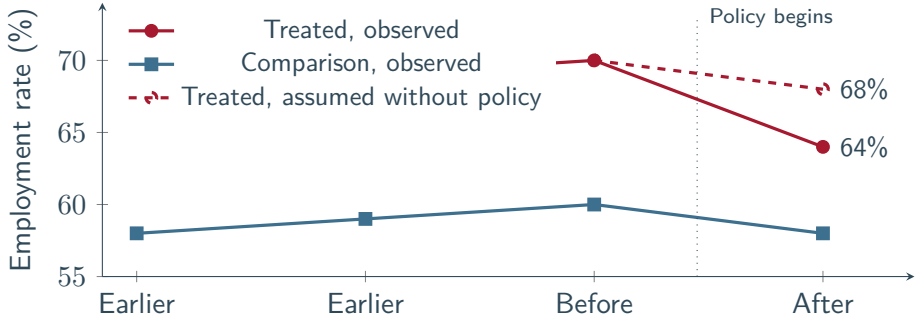
Group	Before	After	Change
Treated	70%	64%	-6 pp
Comparison	60%	58%	-2 pp

$$\text{DiD} = \underbrace{(64 - 70)}_{\text{treated change}} - \underbrace{(58 - 60)}_{\text{comparison change}} = -4 \text{ pp}$$

If, without the policy, the treated group would also have changed by -2 pp, its counterfactual after rate is $70 - 2 = 68\%$. The observed 64% is 4 percentage points below that assumed counterfactual.

Source: Two-group, two-period classroom example. A causal interpretation requires the assumptions on slide 12.

Parallel trends supply the unobserved path



Source: Invented illustration. The dashed segment imposes parallel untreated trends. Similar earlier trends support plausibility but cannot prove the missing path.

When the comparison can mislead

Parallel trends: absent the treatment, the groups would have had the same average change in the outcome.

- No anticipation: behaviour should not already respond before the chosen treatment date.
- No spillovers: the treatment should not change the comparison group's outcome.
- Other shocks, changing populations, or different measurement can generate different trends.

In trade, shared supply chains and retaliation can affect both groups. Export exposure may also reflect earlier industrial decline. Pre-policy plots, institutions, and alternative comparisons help assess the design. A good-looking pre-trend plot alone cannot establish causality.

Source: Roth et al. (2023), survey of differences-in-differences; see linked reading. The paper's calibrated welfare comparison is not a DiD estimate.

A clear claim makes the evidence useful

1. Fix the unit, outcome, policy date, and comparison. Explain the economic channel.
2. Plot the underlying data and check definitions, denominators, and earlier trends.
3. State whether the evidence is descriptive or supports a conditional causal interpretation.
4. Connect the measured outcome to household income, prices, assets, and adjustment.

A prompt to copy into Codex

Audit my proposed comparison for [Canadian case]. List the identifying assumption, spillovers, anticipation, and other shocks. Plot pre-policy outcomes. Separate verified facts from assumptions. If the design is weak, explain what a useful descriptive analysis can establish.

Use this toolkit for the final project. It will not be tested on Midterm 2.

Readings, data, and project resources

- Ma and Ramondo, *Comparative Advantage and Economies of Scale under Uncertainty*, October 2026 working paper. Model pp. 6–15; Tables 6, 7, and 10.
- Statistics Canada 14-10-0023-01: annual industry employment. The course starter preserves the source definitions and units.
- Roth, Sant'Anna, Bilinski, and Poe (2023), *What's Trending in Difference-in-Differences? A Synthesis of the Recent Econometrics Literature*.
- Canadian specialization project handbook and starter data ZIP.
- Podcast, transcript, chapters, and previous-course review.

Scope: this new paper and causal-reasoning workshop support the final project. They add no Midterm 2 test material.